

# Pratyush Singh

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## SUMMARY

Computer Science Engineering student specializing in Cloud Computing and Automation, with hands-on experience in cloud platforms, containerization, and deploying production web applications on the cloud. Proficient in AWS, Docker, Git, and GitHub, with a strong backend foundation in Node.js, REST APIs, and PostgreSQL. Experienced in building and deploying secure, multi-tenant cloud applications with authentication, role-based access control, and real-time services, along with strong problem-solving skills.

## EDUCATION

### VIT Bhopal University

*B.Tech in Computer Science and Engineering (Cloud Computing and Automation)*

*CGPA: 8.00/10*

Bhopal, Madhya Pradesh

*May 2027*

### Bhavan's Kesari Devi Kanoria Vidya Mandir

*Class-X: 92.5%*

*Class-XII: 73.5%*

Sonebhadra, Uttar Pradesh

*2021*

*2023*

## TECHNICAL SKILLS

**Languages :** Python, C++, JavaScript, TypeScript, SQL, HTML, CSS

**Frameworks :** React.js, Next.js, Node.js, Express.js, Flask, FastAPI

**Databases :** PostgreSQL, MongoDB, MySQL, SQLite

**Cloud & DevOps :** AWS (EC2), Docker, Git, GitHub, Cloud Deployment (Vercel, Neon)

## PROJECTS

**WorkNest** | *Next.js, TypeScript, Node.js, Express, PostgreSQL, Prisma, Socket.IO, JWT*

Oct 2025 – Present

- Built and deployed a multi-tenant cloud SaaS workspace unifying HR, project delivery, and a client portal for service companies, exposing 85+ REST API endpoints across 18 Express modules on a 23-model PostgreSQL schema via Prisma.
- Implemented JWT authentication in httpOnly cookies with bcrypt hashing and role-based access control (RBAC) for 4 roles (Admin, Manager, Employee, Client), isolating tenants by scoping every query to a company ID.
- Secured onboarding with single-use SHA-256-hashed invite and password-reset tokens, brute-force rate limiting (15-minute lockout after 8 failed sign-ins), Zod schema validation, and searchable audit logs.
- Built a real-time layer with Socket.IO over cookie-authenticated WebSockets and isolated per-project rooms, live-syncing a drag-and-drop Kanban board, activity feeds, attendance, and notifications.
- Developed geofenced attendance (Haversine distance), leave-approval workflows, and an AI assistant using LLM tool calling (Gemini/Groq); Node.js/Express backend deployed on AWS EC2, with Vercel frontend and Neon serverless PostgreSQL.

**Wardrobe AI** | *Next.js, React, TypeScript, Gemini API, Claude API, SQLite, Zod, Tailwind CSS*

Oct 2026

- Developed an AI-powered outfit assistant that analyzes an uploaded outfit photo with a multimodal vision LLM and returns a 0–100 suitability score across 4 dimensions: formality, coordination, weather, and event context.
- Engineered a provider-agnostic cloud AI layer integrating Google Gemini and Anthropic Claude APIs with Zod-validated structured JSON output, ensuring type-safe responses and consistent error handling across both providers.
- Built a digital closet with automatic AI tagging of category, color, fabric, season, and occasion, plus outfit recommendations with a rule-based fallback that keeps suggestions working when the AI is unavailable.
- Designed 5 REST API routes with the Next.js App Router and SQLite, integrated live weather from the cloud-based Open-Meteo API, and added client-side image compression to keep photo uploads under the 5 MB limit.

## CERTIFICATIONS

NPTEL: Introduction to Internet of Things Certificate

Microsoft: Azure Data Fundamentals Certificate

## ACHIEVEMENTS

- Achieved a 5-Star rating in C++ on HackerRank, demonstrating strong proficiency in C++ programming, Data Structures, Algorithms, and Problem Solving.
- Solved 300+ coding problems across LeetCode and other programming platforms and achieved a LeetCode ranking under 500,000, demonstrating strong problem-solving skills in Data Structures and Algorithms.